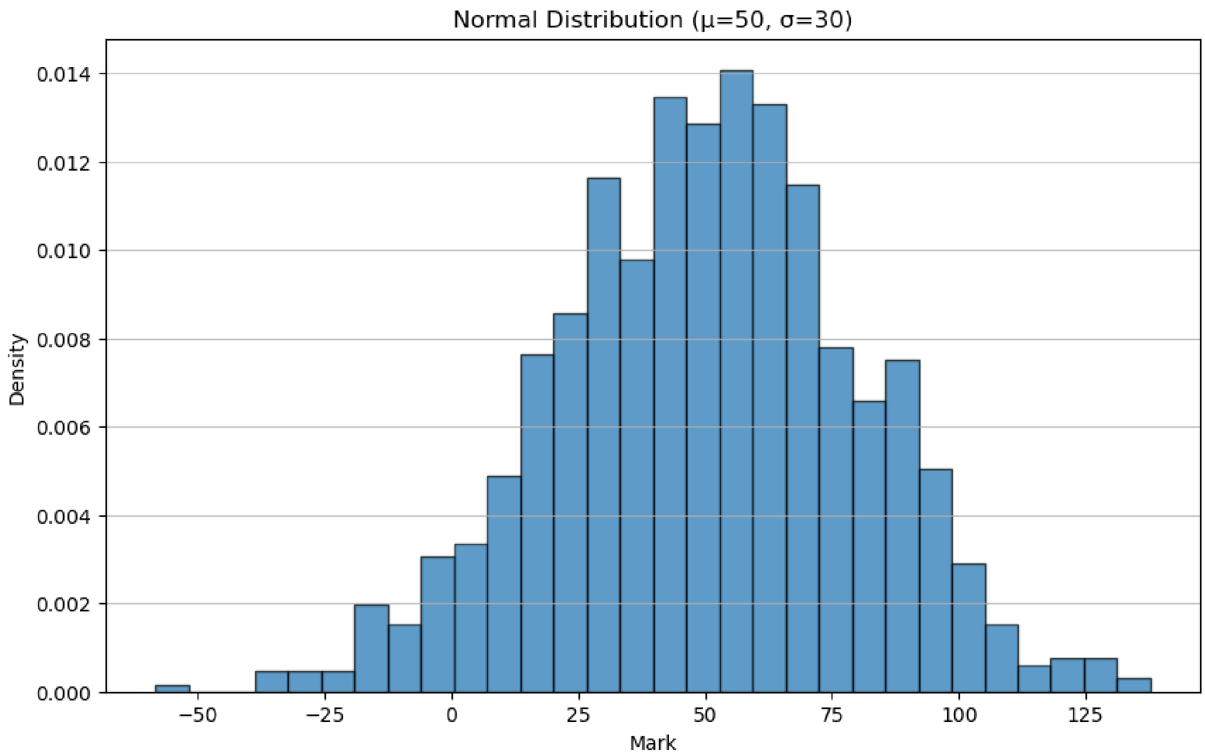
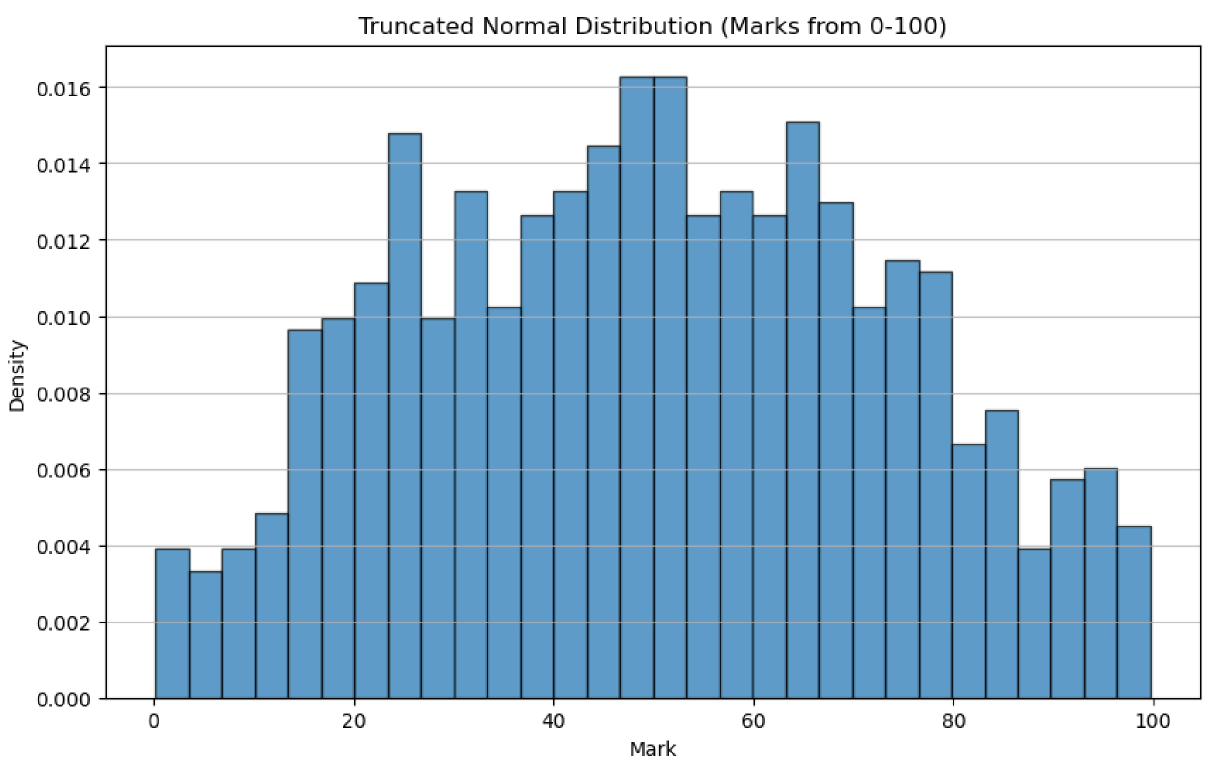
In [4]:

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import truncnorm

mu = 50
sigma = 30
size = 1000

normal_marks = np.random.normal(loc=mu, scale=sigma, size=size)

plt.figure(figsize=(10, 6))
plt.hist(normal_marks, bins=30, density=True, edgecolor='black', alpha=0.7)
plt.title(f'Normal Distribution (μ={mu}, σ={sigma})')
plt.xlabel('Mark')
plt.ylabel('Density')
plt.grid(axis='y', alpha=0.75)
plt.show()

lower_bound = 0
upper_bound = 100

a = (lower_bound - mu) / sigma
b = (upper_bound - mu) / sigma

truncated_marks = truncnorm.rvs(a, b, loc=mu, scale=sigma, size=size)

plt.figure(figsize=(10, 6))
plt.hist(truncated_marks, bins=30, density=True, edgecolor='black', alpha=0.7)
plt.title('Truncated Normal Distribution (Marks from 0-100)')
plt.xlabel('Mark')
plt.ylabel('Density')
plt.grid(axis='y', alpha=0.75)
plt.show()

calculated_mean = np.mean(truncated_marks)
calculated_std = np.std(truncated_marks)

print(f"Original Mean (μ): {mu}")
print(f"Original Standard Deviation (σ): {sigma}")
print("-" * 30)
print(f"Calculated Mean (μ) of Truncated Data: {calculated_mean:.4f}")
print(f"Calculated Standard Deviation (σ) of Truncated Data: {calculated_std:.4f}")

plt.figure(figsize=(10, 6))
plt.hist(z_scores, bins=30, density=True, edgecolor='black', alpha=0.7)
plt.title('Distribution of Z-scores')
plt.xlabel('Z-score')
plt.ylabel('Density')
plt.grid(axis='y', alpha=0.75)
plt.show()

print(f"Mean of Z-scores: {np.mean(z_scores):.4f}")
print(f"Standard Deviation of Z-scores: {np.std(z_scores):.4f}")
```

```
plt.figure(figsize=(10, 6))
plt.scatter(truncated_marks, z_scores, s=10, alpha=0.5)
plt.title('Raw Mark vs. Z-score')
plt.xlabel('Raw Mark')
plt.ylabel('Z-score')
plt.grid(True)
plt.show()
```





```
Original Mean (μ): 50
Original Standard Deviation (σ): 30
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Calculated Mean (μ) of Truncated Data: 49.9533
Calculated Standard Deviation (σ) of Truncated Data: 23.6295
```
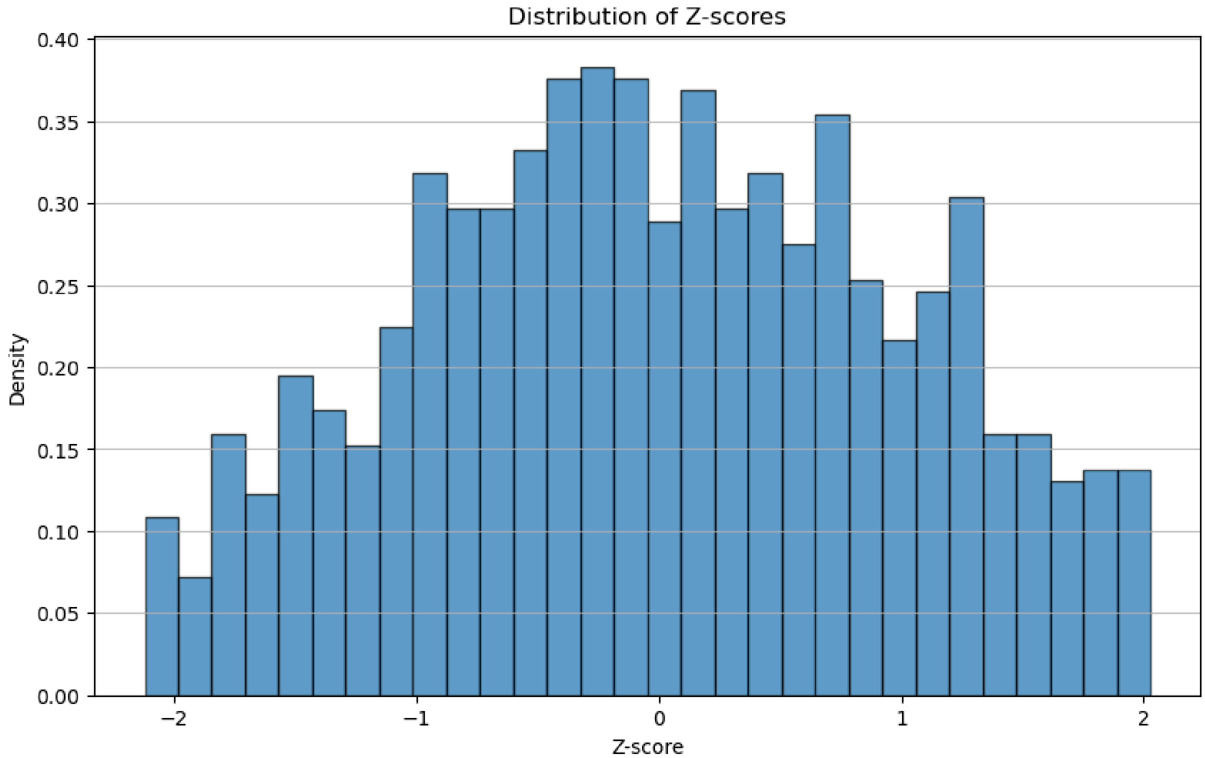
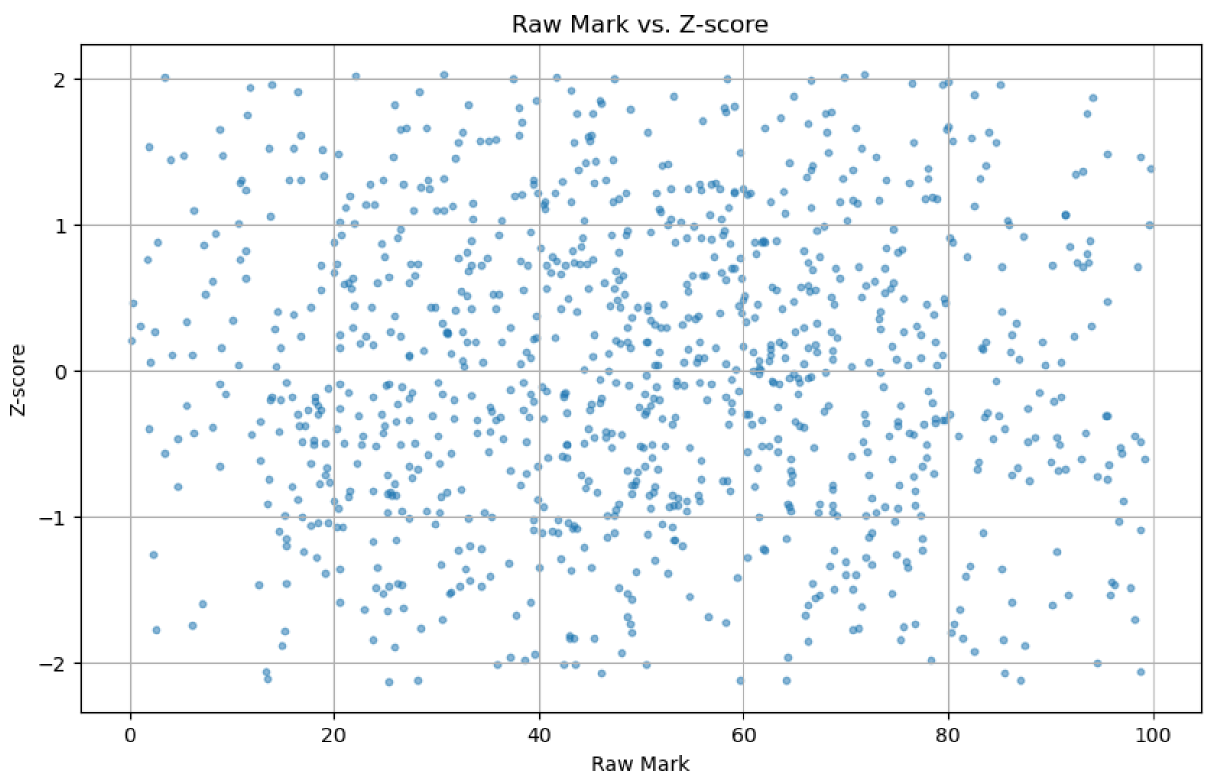




```
Mean of Z-scores: -0.0000
Standard Deviation of Z-scores: 1.0000
```



In [ ]: